

Notes_0305

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1 Methods

1.1 SWIM for Rational Function

1.1.1 Activation Function

A rational function like following is used as activation function:

$$\sigma(x) = \frac{C - m \prod_{i=1}^3 (x - a_i)^2}{C + m \prod_{i=1}^3 (x - a_i)^2} + 1 \quad (1)$$

where $C, m > 0$ is some constant value. a_i also some constant value. The function has the following properties: It has 3 peaks; at all these 3 peaks, it reaches its highest value 2; The function goes to zero at infinity. Here, C, m together control how spiky the function is. In the following experiments, we fix $C = 10, m = 0.01$. We also fix $a_1 = 1, a_2 = 0, a_3 = -1$.

1.1.2 Parametrization of Activation Function

For each neuron, we introduce weight $\mathbf{w}$ and bias b :

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, b$$

We then plugin $\mathbf{x}$ into the activation function using linear transformation.

$$\sigma(\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b) \quad (2)$$

This is obviously equivalent to:

$$\sigma(x_1, x_2) = \frac{C - m \prod_{i=1}^3 (w_1 x_1 + w_2 x_2 + b - a_i)^2}{C + m \prod_{i=1}^3 (w_1 x_1 + w_2 x_2 + b - a_i)^2} + 1 \quad (3)$$

In the perspective of rational function, w_1, w_2 and b now become the coefficients of the polynomials. This is how we parameterize the activation function.

1.1.3 Sampling Strategy

Assume the objective dynamic we want to optimize be:

$$y = f(\mathbf{x}) : \mathbb{R}^2 \rightarrow \mathbb{R} \quad (4)$$

For each neuron, we need to find the appropriate $\mathbf{w}$ and b , to do that, we randomly sample 3 data points $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, the probability that we sample these 3 points is proportional to the summation of the corresponding 3 y values. In other words,

$$(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \sim p \quad (5)$$

$$p \propto \sum_{i=1}^3 f(\mathbf{x}_i) \quad (6)$$

Then we find $\mathbf{w}$ and b such that:

$$\mathbf{w}^T \mathbf{x}_i + b = a_i \quad i \in \{1, 2, 3\}$$

This ensures the 3 points in the set we sampled align to the peak position of our rational function. This can be solved by:

$$\begin{bmatrix} \mathbf{w} \\ b \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^T & 1 \\ \mathbf{x}_2^T & 1 \\ \mathbf{x}_3^T & 1 \end{bmatrix}^{-1} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad (7)$$

Assume that we have M neurons, after doing this for all neurons, we get the weights $\mathbf{W} \in \mathbb{R}^{2 \times M}$, and biases $\mathbf{b} \in \mathbb{R}^{1 \times M}$. The single layer SWIM neuron network can be written as:

$$SWIM(\mathbf{x}) = \mathbf{w}'^T \sigma(\mathbf{W}^T \mathbf{x} + \mathbf{b}) + b' \quad (8)$$

Where $\mathbf{w}' \in \mathbb{R}^M$ and $b \in \mathbb{R}$ are found by linear solution after sampling $\mathbf{W}$ and $\mathbf{b}$ such that the l2 norm is minimized.

$$\mathbf{w}', b' = \operatorname{argmin}(f(\mathbf{X}) - SWIM(\mathbf{X}))^T (f(\mathbf{X}) - SWIM(\mathbf{X})) \quad (9)$$

1.1.4 Performance

The following figures show the performance of SWIM-rational comparing with traditional SWIMRelu and SWIM tanh when approximating different dynamics. Here I list 3 dynamics: KdV_sine, gaussian, 2d_tanh.

1.2 SWIM for ReLu 2nd gradient